Aashish Gupta

VICO Postdoctoral Fellow at the Department of Astronomy, University of Virginia, USA

aashishgupta@virginia.edu ◦ aashishgpta.github.io ◦ 0000-0002-9959-1933 ◦ ADS Library

RESEARCH INTERESTS

I study how stars and planets form, primarily using observations from a range of telescopes. My recent work focuses on the interactions between star-forming environments and planet-forming disks, and on how they alter the material available to build planets.

HIGHLIGHTS

- 16 (+8 submitted) refereed research publications, including 8 (+2 submitted) with major contributions (more details).
- PI on 8 approved programs across ALMA (105.8 h), GBT (21.4 h), SMA (8 h), and VLT (3.7 h) (more details).
- 28 research talks given, including 14 invited talks (more details).
- Mentored 6 students on astronomy-related projects, ranging from a high school student to graduate students (more details).
- Founded and running "The Big Science Show", a monthly public science outreach event in Charlottesville with 150+ subscribers.
- Conceived and chairing the organizing committee for the "DISCO: Disks in Context", a star and planet formation conference with 60+ participants.

EDUCATION

Doctor of Philosophy (Physics) — *2021 – 2024*
European Southern Observatory (ESO) and
Ludwig Maximilian University (LMU), Germany — *Magna cum laude*

Master of Science (Astronomy) — *2019 – 2021*
National Central University (NCU), Taiwan

Bachelor of Technology (Mechanical Engineering) — *2015 – 2019*
Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur, India

PROFESSIONAL APPOINTMENTS

(Upcoming) **Lecturer**, University of Virginia (UVA), USA — *Fall 2026*

VICO postdoctoral fellow, University of Virginia (UVA), USA — *2025 – present*
Advisor: Dr. Ilse Cleeves
I am leading multiple research projects to characterize the impact of star-forming environments on planet-forming disks.

Doctoral researcher, European Southern Observatory (ESO), Germany — *2021 – 2024*
Advisor: Dr. Anna Miotello
I demonstrated that planet-forming disks often interact with surrounding molecular clouds, and such clouds can drastically enhance their mass-budget to form planets.

Graduate researcher, National Central University (NCU), Taiwan — *2019 – 2021*
Advisor: Prof. Chen, Wen-Ping
I diagnosed the distribution of the young stars with respect to their natal molecular clouds.

Summer intern, Academia Sinica Institute of Astronomy and Astrophysics (ASIAA), Taiwan *2020*
Advisor: Dr. Yen, Hsi-Wei
I investigated influence of magnetic fields on gas kinematics within protostellar envelopes.

Project-based internship, Aryabhatta Research Institute of Observational Sciences (ARIES), India *2018*
Advisor: Dr. Jeewan Pandey
I developed a pipeline to detect and analyze stellar flares.

Summer intern, Aryabhatta Research Institute of Observational Sciences (ARIES), India *2017*
Advisor: Dr. Sneh Lata
I developed a pipeline to analyse time series data of RR Lyrae Variables.

Undergraduate researcher, Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur, India *2016*
Advisor: Prof. Puneet Tandon
I assisted in the design and fabrication of a 3D printing machine.

SKILLS

Programming	Python, C/C++, Octave, HTML, Shell scripting, LaTeX
Languages	Hindi (Native), English (Proficient, IELTS score 8), German (Basic, A1), Spanish (Basic, A1), Mandarin Chinese (Basic speaking and listening)

OBSERVING EXPERIENCE

PI Projects:

1. ALMA project "The invisible majority: Testing CI as a tracer for CO-dark streamers" *(6.4 h on 12 m array, 24.9 h on ACA)* *2026.1.00679.S*
2. ALMA project "Zooming Out and Peering In: The First Statistical Study of Infall onto Class II Disks" *(12.3 h on 12 m array)* *2025.1.00328.S*
3. ALMA project "Paradigm Shift or Rare Events: Testing the Frequency of Late Infall Using Herbig Disks" *(16.2 h on ACA, 25.3 h on TP)* *2025.1.00197.S*
4. ALMA project "Is infall of material misaligning planet-forming disks?" *(8.7 h on 12 m array)* *2025.1.00803.S*
5. ALMA project "Zooming out: Disentangling the origins of large-scale structures around planet-forming disks" *(11.8 h on ACA)* *2024.1.00524.S*
6. GBT project "Characterizing gas reservoirs feeding planet-forming disks" *(21.4 h)* *GBT26A-579*
7. SMA project "In or Out: Spectral-Line Survey of a Streamer" *(8 h)* *2025A-S031*
8. VLT project "Exploring effects of infall of material onto Class II systems" *(3.7 h on X-Shooter)* *113.26GY*

Involvement in Large Programs:

1. Disk-Exoplanet C/Onnection (DEC/O) *2022.1.00875.L*
   138.8 h on ALMA, PI: Ilse Cleeves
   I am leading the efforts on characterizing the large-scale structures observed in DEC/O. I was also involved in the development of imaging pipeline, submitting follow-up proposals (2 accepted), and giving feedback on manuscripts.
2. Chemistry of Herbig Environments and their Exoplanet Relationships (CHEER) *2024.1.01001.L*

89.7 h on ALMA, PI: Jamila Pegues
I am co-leading the efforts on characterizing the large-scale structures observed in CHEER. I am also involved in the development of imaging pipeline and submitting follow-up proposals (2 accepted).

3. The JCMT Transient Survey *M16AL001*
200+ h on JCMT, PIs: Gregory Herczeg, Doug Johnstone
I lead the analysis of coadd image for the Ophiuchus region. I also helped with literature review of interesting sources.

CoI Projects:

CoI on 14 approved programs (700+ h): 13 ALMA, 2 JWST, 2 VLT, 2 APEX, 1 SMA, and 1 SMT.

Hands-on Observing:

Assisted with optical observations at ARIES, India and Lulin Observatory, Taiwan.

SOFTWARE DEVELOPMENT

TIPSY – Trajectory of Infalling Particles in Streamers around Young stars *2024*

I developed TIPSY, the first-of-its-kind open-source code for characterizing “streamers”, i.e., trails of gas falling onto young stellar systems. TIPSY fits the molecular-line observations of streamers in 3D position–position–velocity space and constrains dynamical parameters like energy, angular momenta, infall timescale, etc. TIPSY-based analysis has been central to over 8 refreed publications so far (visit the GitHub repository for more details).

MENTORING EXPERIENCE

1. Supervising Mason Mabry for Interconnected Cosmos Undergraduate Research Program at UVA *2026 – present*
2. Co-supervising Noshin Yesmin for her Ph.D. project at UVA *2026 – present*
3. Supervised Andres Zuleta for “Cool Frontiers: Exploring Dust and Ice in the Cosmos” at UVA *2025*
4. Co-supervised Lauren Mason for the ESO Summer Research Programme *2024*
5. Supervised Jayden Burg, a high school student, for one week internship (Schülerpraktikum) *2024*
6. Co-supervised Alessandro Ruzza for the ESO Summer Research Programme *2023*
7. Student guide for the first year students at IIITDMJ *2016*

TEACHING EXPERIENCE

1. *(Upcoming)* ASTR 3150 “The Interstellar Medium: From Hydrogen to Humans" *Fall 2026*
2. Lecture on planet formation for ASTR 1610: “Introduction to Astronomical Research” at UVA *Spring 2026*
3. Academic Helper for Physics and Mathematics at IIITDMJ Counseling Services *2016 – 2018*

ACADEMIC SERVICE

1. Chair of the organizing committee for the “DISCO: Disks in Context”, a star and planet formation conference *2025 – present*
2. Chair of Community Building Committee of the UVA Postdoc Association *2025 – present*
3. Local Organizing Committee member for “Cool Frontiers: Exploring Dust and Ice in the Cosmos” at UVA *2025*
4. Local Organizing Committee member for DECO All Team meeting at ESO, Germany *2024*

5. Local Organizing Committee member for "New Heights in Planet Formation" at ESO, Germany *2024*
6. Scientific Assistant for ESO Observing Programmes Committee *2024*
7. Member of selection committee for ESO Summer Research Programme *2024*
8. Co-organizer of Wine & Cheese seminars at ESO, Germany *2023 – 2024*
9. Chaired for Hypatia Colloquium at ESO, Germany *2023*
10. Refereed for Science Advances and The Astrophysical Journal

OUTREACH

1. Founding organizer for "The Big Science Show". Every month, we host two local scientists, from any field of research, to present their topic of interest to the community. *2026 – present*
2. Co-organizer of "Go Explore the Universe" summer camp at Charlottesville, USA *2026*
3. Outreach talk at Astronomy on Tap, Charlottesville, USA *2026*
4. Public night talk at McCormick Observatory at Charlottesville, USA *2026*
5. Co-organizer of "Girls Exploring the Universe" summer camp at Charlottesville, USA *2025*
6. Co-organizer of "Astronomy for Non-astronomers" monthly talk series at ESO, Germany *2024*
7. Coordinator of The Astronomy and Physics Society of IIITDM Jabalpur, India. We organized events like telescope nights, exhibitions, quizzes, etc. to promote astronomy among a variety of audiences. *2016 – 2019*
8. E-Astronomer at RAD@home Astronomy Collaboratory, an Indian citizen-science research programme *2018 – 2019*

AWARDS AND GRANTS

Total awards and grants worth ∼$ 441,730.

1. **$ 218,000** – VICO postdoctoral fellowship at UVA, USA *2025 – 2028*
2. **$ 6,500** – Global Initiative Grant from Center for Global Inquiry and Innovation at UVA, USA for organizing "DISCO: Disks in Context" conference *2026*
3. **€ 172,550 (∼$ 200,000)** – DFG funding for Ph.D. at ESO, Germany *2021 – 2024*
4. **€ 4,000 (∼$ 4,500)** – ERC DUSTBUSTERS funding for a two-month secondment at the University of Hawaii, USA *2022*
5. **NT$ 361,100 (∼$ 11,180)** – MOST scholarship (tuition fees waiver and support of living expenses) for master's program at NCU, Taiwan *2019 – 2021*
6. **NT$ 20,000 (∼$ 620)** – Award for academic performance at NCU, Taiwan *2020*
7. Poster presentation award, Annual Meeting of the Physical Society of Taiwan *2020*
8. **₹ 60,000 (∼$ 630)** – ARIES funding for semester-long internship, Nainital, India *2018*
9. Certificate of Merit for academic excellence at IIITDM Jabalpur, India *2017*

PARTICIPATION IN MEETINGS

Invited talks:

1. Astronomy Department Talk, University of Michigan, Ann Arbor, USA *July 2026*
2. Astronomy Seminar, Tata Institute of Fundamental Research, Mumbai, India *February 2026*
3. Astronomy Seminar, University of Connecticut, Storrs, USA *October 2025*
4. Star and Exoplanet Seminar, Yale University, New Haven, USA *October 2025*
5. Center for Astrophysics | Harvard & Smithsonian, Cambridge, USA *June 2025*

6. European Astronomical Society Annual Meeting (EAS), Cork, Ireland *June 2025*
7. StarPlan seminar, University of Copenhagen, Copenhagen, Denmark *December 2024*
8. Astronomy seminar, Universidad Diego Portales, Santiago, Chile *September 2024*
9. Astronomy seminar, University of Exeter, Exeter, UK *February 2024*
10. Core2disk III workshop, Paris, France *October 2023*
11. Origins seminar at Chalmers University, Gothenburg, Sweden *August 2023*
12. ESO star and planet formation seminar, Garching, Germany *February 2023*
13. ECOGAL seminar, Online *February 2023*
14. MPE-CAS journal club on star and planet formation, Garching, Germany *July 2022*

Contributed talks:

1. Emerging Researchers in Exoplanet Science Symposium XI, Ann Arbor, Michigan *July 2026*
2. MIAPbP workshop on "Planet Formation across various Environments and Epochs", Garching, Germany *May 2026*
3. ESO Wine & Cheese Seminar *May 2026*
4. UVA Postdoc Research Symposium *May 2026*
5. VICO Fall Workshop 2025, Charlottesville, USA *December 2025*
6. The Chesapeake Bay Area Exoplanet Meeting #12, Laurel, USA *May 2025*
7. VICO Spring Workshop 2025, Charlottesville, USA *May 2025*
8. UVA Postdoc Research Symposium *October 2024*
9. Born in Fire: Eruptive Stars and Planet Formation, Santiago, Chile *September 2024*
10. New Heights in Planet Formation, Garching, Germany *July 2024*
11. ESO Wine & Cheese Seminar *January 2024*
12. European Astronomical Society Annual Meeting (EAS), Krakow, Poland *July 2023*
13. Meeting of ALMA Young Astronomers (MAYA), Online *March 2023*
14. Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan *September 2020*

Posters:

1. European Astronomical Society Annual Meeting (EAS), Cork, Ireland *June 2025*
2. Origins of Solar Systems (Gordon Research Conference), South Hadley, Massachusetts, USA *June 2025*
3. Inter+Stellar: Harnessing the Intersection Between Stars and the Interstellar Medium, Baltimore, USA *May 2025*
4. Star@Lyon conference, Lyon, France *June 2023*
5. Protostars and Planets VII, Kyoto, Japan *April 2023*
6. Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan *September 2020*
7. Annual Meeting of the Physical Society of Taiwan (TPS), Pingtung, Taiwan *February 2020*
8. Exploring the Universe: Near Earth space science to extragalactic astronomy (EXPUNIV2018), Kolkata, India *November 2018*

PUBLICATIONS

Total citations: 301, first author citations: 112. h index = 10.

Major-Contribution Publications:

1. *(Submitted) DECO x: Evidence of widespread interactions between planet-forming disks and their surrounding clouds*

A. Gupta, T. J. Haworth, A. Miotello, et al..

2. *(Submitted) Disk Asymmetries Grow with Stellar Mass: Evidence from 67 Planet-forming Disks*
   A. Zuleta, **A. Gupta**, P. Curone, et al.
3. *Angular Momentum of Planet-Forming Disks: Implications for Infall Driven Misalignments*
   A. Gupta, C. Longarini, L. I. Cleeves, et al. (2026), accepted in A&A. 10.48550/arXiv.2607.23741
4. *A tale of three tails: A misaligned streamer and mysterious structures around [BHB2007]-1*
   A. Gupta, A. S. Hales, L. I. Cleeves, et al. (2026), ApJ, 998, 105. 10.3847/1538-4357/ae2477
5. *Discovery of an Accretion Streamer and a Slow Wide-angle Outflow around FU Orionis*
   A. S. Hales, **A. Gupta**, D. Ruíz-Rodríguez, et al. (2024), ApJ, 966, 96. 10.3847/1538-4357/ad31a1
6. *TIPSY: Trajectory of Infalling Particles in Streamers around Young Stars. Dynamical analysis of streamers around S CrA and HL Tau*
   A. Gupta, A. Miotello, J. P. Williams, et al. (2024), A&A, 683, A133. 10.1051/0004-6361/202348007
7. *A dusty streamer infalling onto the disk of a Class I protostar: ALMA dual-band constraints on grain properties and mass infall rate*
   L. Cacciapuoti, E. Macías, **A. Gupta**, et al. (2024), A&A, 682, A61. 10.1051/0004-6361/202347486
8. *Reflections on nebulae around young stars: A systematic search for late-stage infall of material onto Class II disks*
   A. Gupta, A. Miotello, C. F. Manara, et al. (2023), A&A, 670, L8. 10.1051/0004-6361/202245254
9. *Effects of Magnetic Field Orientations in Dense Cores on Gas Kinematics in Protostellar Envelopes*
   A. Gupta, H.-W. Yen, P. Koch, et al. (2022), ApJ, 930, 67. 10.3847/1538-4357/ac63bc
10. *Interplay between Young Stars and Molecular Clouds in the Ophiuchus Star-forming Complex*
    A. Gupta, W.-P. Chen (2022), AJ, 163, 233. 10.3847/1538-3881/ac5cc8

Other Publications:

1. *JWST-DECO: The impact of accretion on mid-infrared observable water in planet-forming disks*
   J. K. Calahan, T. Dziire, … **A. Gupta**, et al. (2026), ApJ, 1004, 213. 10.3847/1538-4357/ae707a
2. *The environmental imprint on molecular layering in the dusty streamer of M512*
   M. De Simone, L. Cacciapuoti, … **A. Gupta**, et al. (2026), A&A, 709, L10. 10.1051/0004-6361/202659631
3. *Spatially correlated stellar accretion in the Lupus star-forming region: Evidence for ongoing infall from the interstellar medium*
   A. J. Winter, M. Benisty, … **A. Gupta**, et al. (2024), A&A, 691, A169. 10.1051/0004-6361/202452120
4. *Anatomy of the Class I protostar L1489 IRS with NOEMA: disk, streamers, outflow(s) and bubbles at 3 mm*
   M. Tanious, R. Le Gal, … **A. Gupta**, et al. (2024), A&A, 687, A92. 10.1051/0004-6361/202348785
5. *The VLT MUSE NFM view of outflows and externally photoevaporating discs near the Orion Bar*
   T. J. Haworth, M. Reiter, … **A. Gupta**, et al. (2023), MNRAS, 525, 4129. 10.1093/mnras/stad2581
6. *The JCMT Transient Survey: Four-year summary of monitoring the submillimeter variability of protostars*
   Y.-H. Lee, D. Johnstone, … **A. Gupta**, et al. (2021), ApJ, 920, 119. 10.3847/1538-4357/ac1679
7. *No impact of core-scale magnetic field, turbulence, or velocity gradient on sizes of protostellar disks in Orion A*
   H.-W. Yen, B. Zhao, P. M. Koch, **A. Gupta** (2021), ApJ, 916, 97. 10.3847/1538-4357/ac0723
8. *VR CCD photometry of variable stars in the globular cluster NGC 4147*
   S. Lata, A. K. Pandey, … **A. Gupta**, et al. (2019), AJ, 158, 51. 10.3847/1538-3881/ab22a6